

SITE-PROBE ACCURACY IS NOT A HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level MRI harmonization is often scored by how poorly a frozen site classifier recognizes the acquisition site of harmonized images. We audit this practice on raw ABIDE I T1-weighted MRI (1,112 subjects, 17 sites), using ten outputs from seven harmonization families as test cases. With subject-level paired intervals and valid controls we find that: (i) five probes trained identically, all accurate on raw images, assigned the same CycleGAN outputs site balanced accuracies from 0.13 to 0.59; (ii) the lowest accuracy, at chance, came from the most destructive output; (iii) diffusion translation changed intensities least but moved them no closer to the target site; and (iv) probes trained on harmonized images recovered the source site almost as well as from raw images (0.87–0.91). Head silhouettes alone identified sites (0.73–0.86), and an intensity-only probe showed that histogram matching removed intensity site information while CycleGAN and diffusion retained it. We propose a reporting checklist.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Acquisition differences between sites confound pooled MRI analyses and degrade models on unseen scanners [1, 2]. Image-level harmonization translates scans toward a common appearance. Travelling subjects [3] or paired protocols [4] allow direct checks but cover few subjects and scanners, so methods applied to large retrospective cohorts are judged indirectly [5]. A common proxy is a *site probe*: a classifier trained to recognize the site on unharmonized images and applied, frozen, to harmonized ones [2, 6]. Lower accuracy is read as removal of site information and is paired with similarity to the input, such as PSNR, and with intensity-distribution distances.

Each proxy measures less than it appears to. A frozen probe reports what one classifier recovers; it can also fail be-

cause outputs are unfamiliar, damaged or noisy. Removing an attribute until a fixed adversary fails does not stop a newly trained classifier from recovering it [7], and for harmonized features site classifiers retrained after ComBat still predicted site above chance [8]. Similarity to the input rewards doing nothing, and a small input-to-output distance says nothing about the target site.

We audit these practices on outputs of seven harmonization families. Contributions: (1) a subject-level protocol with paired uncertainty, valid probe controls, separate reporting of target-site subjects, and target alignment measured apart from change; (2) evidence that probe verdicts depend on the checkpoint, that the lowest site accuracy can signal destruction, and that small change does not imply alignment; (3) a test of removed versus hidden site information using probes trained on harmonized images, with geometry-only and intensity-only controls; and (4) a reporting checklist.¹ We neither propose nor rank harmonizers.

2. DATA AND AUDITED OUTPUTS

Cohort. Raw ABIDE I T1-weighted scans [9] of 1,112 subjects from 17 sites are split by subject, stratified by site, into 890 training, 113 validation and 109 test subjects. Volumes are reoriented, clipped to foreground percentiles $[p_1, p_{99}]$ and scaled to $[0, 1]$. One axial slice per subject is chosen by foreground area, cropped with a dilated head mask and resized to 256×256 . NYU, the largest site, is the target; 19 test subjects are from NYU and 90 from 16 source sites. Four test subjects have exactly tied candidate slices, and one tie resolved differently on another machine, so slices are frozen and verified for every output.

Audited outputs. Earlier benchmark outputs were evaluated without retraining: NeuroCombat [10, 1] per pixel, fit on the test cohort itself (transductive); histogram matching of foreground quantiles to a pooled 17-site training reference [11]; CycleGAN [12]; StarGAN [13], aggressive and conservative; DLEST [14] after 1,000 and 1,500 steps; condi-

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¹Code, the evaluation protocol with all amendments, and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

tional diffusion image-to-image translation [15, 16] (strength 0.35, 50 DDIM steps); and adapted volumetric HCLD [17]. Learned methods were trained on the training split; generators return NYU inputs unchanged up to numerical precision. The test subjects were inspected during the earlier benchmark, so this is a retrospective reanalysis; audit experiments were fixed in a written protocol before their outcomes were seen, and amendments record what had been observed.

3. AUDIT PROTOCOL

Change and target alignment. PSNR on a fixed $[0, 1]$ scale and normalized cross-correlation (XCorr) compare output with input on whole images; NYU outputs of generators are exact or near-exact identities (PSNR infinite or above 150 dB), so we summarize the 90 source subjects and report NYU separately. *Change* is the per-subject Wasserstein-1 distance W between input and output intensities. *Target alignment* is the Wasserstein-1 distance and KL divergence between foreground intensities and a reference from raw slices of the 147 NYU training subjects, weighted equally; foreground is defined on the raw slice, and ΔW_{NYU} is the output’s distance minus the input’s.

Site probes. Probes are single-channel ResNet-18 networks [18] predicting 17 sites, trained with AdamW (learning rate 3×10^{-4} , batch 64, 500 steps) and random affine augmentation. The *frozen* probe is the earlier benchmark’s checkpoint. *Retrained* probes use the same recipe on random valid training slices (seeds 42, 1–4), selected on validation balanced accuracy (BA) and tested once. A shuffled-label control permutes labels between subjects within each split. To separate removed from hidden site information, *slice probes* are trained with the same recipe on fixed slices of raw images or of one method’s outputs, for three methods chosen in advance to span intensity, adversarial and diffusion families (seeds 1–3). Two controls, added after the first slice-probe results and before their own, restrict the input: slice probes on filled binary head silhouettes (geometry only), and multinomial logistic regression on 50-bin foreground intensity histograms (intensity only), trained on raw slices or on a method’s outputs.

Statistics. Source BA is macro-averaged over the 16 source sites (chance 1/16). Intervals are 95% percentile intervals from 2,000 bootstrap replicates resampling subjects within site, shared across methods and probes so comparisons are paired; they are conditional on fixed checkpoints, and seed variation is reported separately.

4. RESULTS

Probe validity. On raw source images the frozen probe reached BA 0.95 [0.90, 1.00] and the retrained probes 0.81–0.98; the shuffled-label control scored 0.03 [0.01, 0.05]. The probes thus learn site-related structure, but not necessarily

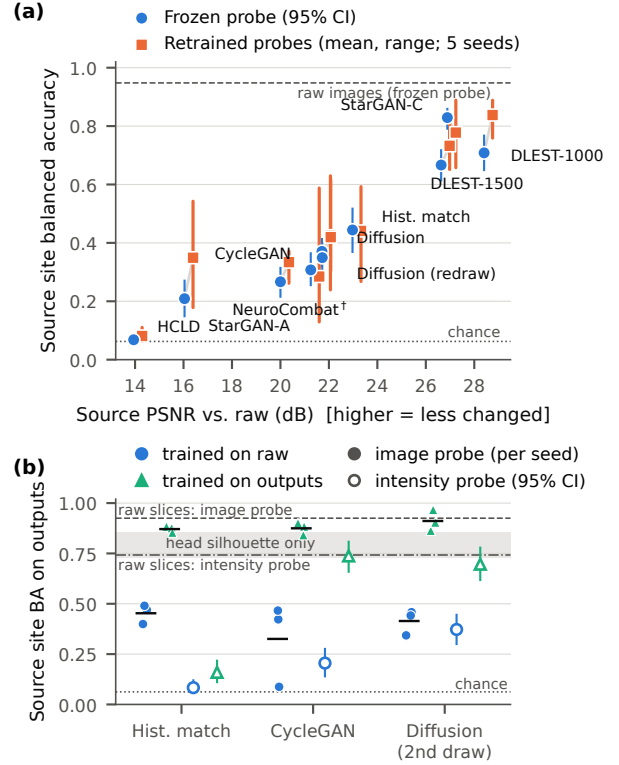


Fig. 1. (a) Site BA versus change from the input. Blue: frozen probe, 95% interval; orange: retrained probes, mean and range over seeds. (b) Probes trained on raw slices (blue) or on each method’s outputs (aqua), tested on that method’s outputs: image slice probes (filled, one per seed, bar at mean) and intensity-histogram probes (hollow, 95% interval). Band: image probes given only head silhouettes.

scanner appearance: slice probes given only head silhouettes reached 0.77, 0.86 and 0.73 (seeds 1–3), and the histogram probe 0.74 [0.65, 0.84].

Verdicts depend on the probe. Identically trained probes assigned very different site BA to the same outputs (Table 1, Fig. 1a): CycleGAN received 0.13–0.59 across seeds, diffusion 0.24–0.63, histogram matching 0.27–0.59 and aggressive StarGAN 0.18–0.54, ranges two to three times the widest bootstrap interval of any single probe (0.16), with the frozen probe inside each range. Ranges were narrower for NeuroCombat (0.26–0.37), HCLD (0.06–0.11) and DLEST 1000 (0.76–0.89). Method orderings agreed moderately (Kendall τ 0.51–0.87 between seeds) and pairs reversed: seed 3 rated aggressive StarGAN above CycleGAN (0.54 vs. 0.20), seed 42 the opposite (0.44 vs. 0.59). Probe strength did not explain this: the probe with the highest raw BA (0.98) gave CycleGAN 0.20.

The lowest site accuracy came from destruction. HCLD reached BA 0.07 [0.06, 0.08] under the frozen probe,

Table 1. Source-subject results ($n = 90$). PSNR and XCorr compare output with input; ΔW_{NYU} : change in Wasserstein-1 distance to the NYU reference (negative is closer). Site BA: frozen probe with 95% interval; retrained probes as mean [min, max] over seeds; image slice probes (mean over seeds) and intensity-histogram probes trained on raw slices $\rightarrow$ on the method’s own outputs. Chance is 0.06. † Fit on the test cohort. ‡ Pooled 17-site reference.

Method	Change and target alignment			Source-site balanced accuracy (probe trained on raw $\rightarrow$ on outputs)			
	PSNR	XCorr	ΔW_{NYU}	Frozen	Retrained	Image probe	Intensity probe
Raw input	—	—	0	0.95	0.90 [0.81, 0.98]	0.92	0.74
NeuroCombat †	20.0	0.940	+0.003	0.27 [0.21, 0.32]	0.33 [0.26, 0.37]	0.36	0.11
Histogram matching ‡	23.0	0.982	-0.053	0.44 [0.37, 0.52]	0.44 [0.27, 0.59]	0.45 $\rightarrow$ 0.87	0.08 $\rightarrow$ 0.16
CycleGAN	21.3	0.977	-0.036	0.31 [0.25, 0.37]	0.28 [0.13, 0.59]	0.33 $\rightarrow$ 0.87	0.21 $\rightarrow$ 0.74
StarGAN (aggr.)	16.0	0.896	-0.019	0.21 [0.15, 0.27]	0.35 [0.18, 0.54]	0.39	0.21
StarGAN (cons.)	26.9	0.991	-0.021	0.83 [0.79, 0.86]	0.78 [0.66, 0.89]	0.82	0.34
DLEST 1500	26.6	0.993	-0.020	0.67 [0.61, 0.72]	0.73 [0.65, 0.82]	0.76	0.42
DLEST 1000	28.4	0.993	-0.009	0.71 [0.65, 0.77]	0.84 [0.76, 0.89]	0.82	0.47
Diffusion img2img	21.7	0.948	-0.001	0.37 [0.32, 0.42]	0.42 [0.24, 0.63]	0.44	0.36
second draw	21.7	0.948	-0.002	0.35 [0.29, 0.40]	0.42 [0.30, 0.62]	0.41 $\rightarrow$ 0.91	0.37 $\rightarrow$ 0.70
Adapted HCLD	13.9	0.706	+0.108	0.07 [0.06, 0.08]	0.08 [0.06, 0.11]	0.07	0.07

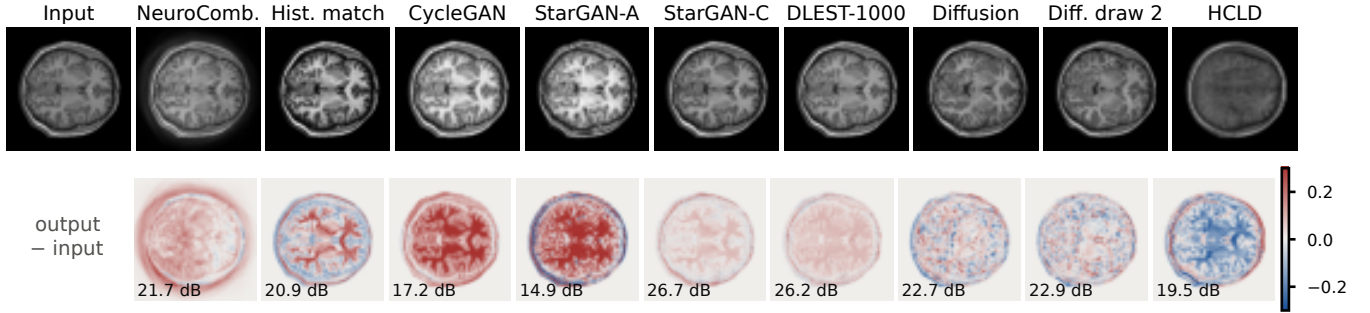


Fig. 2. One source test subject (chosen as the median of PSNR averaged over outputs). Bottom: output minus input, with PSNR. HCLD loses anatomical detail; diffusion adds texture that differs between draws.

at chance, with the lowest PSNR (13.9 dB) and XCorr (0.71) and visible loss of anatomy (Fig. 2), and it moved intensities away from the target ($\Delta W_{\text{NYU}} = +0.108$).

Small change is not target alignment. Diffusion translation changed intensities least ($W = 0.008$) yet moved them no closer to NYU ($\Delta W_{\text{NYU}} = -0.0013 [-0.0023, -0.0004]$) and raised the KL divergence to NYU from 0.30 to 0.83, whereas histogram matching (-0.053) and CycleGAN (-0.036) moved toward the target. Diffusion outputs are also a random draw: resampling the same checkpoint gave images differing from the first draw by a mean foreground absolute difference of 0.085, more than from the input (0.075), at unchanged PSNR (21.71 vs. 21.73 dB). Site BA barely changed between draws (frozen 0.37 vs. 0.35).

Removed or hidden? Slice probes trained on raw slices reached BA 0.92 (0.88–0.97) on raw test slices (shuffled-label control 0.07 [0.03, 0.10]) but only 0.45, 0.33 and 0.41 on histogram-matched, CycleGAN and diffusion outputs, the drop usually reported as harmonization. Probes trained on each method’s outputs recovered the source site from its test outputs with BA 0.87 (0.85–0.88), 0.87 (0.84–0.90) and 0.91 (0.86–0.97) (Fig. 1b); for every seed the intervals of the

two probes were disjoint, also with final-epoch checkpoints. Because geometry alone identifies sites, this shows that site stays recoverable, not through which channel. The intensity-only probe separates the methods: trained on raw slices it fell to 0.08, 0.21 and 0.37 on the three outputs, but trained on the outputs it recovered little from histogram matching (0.16 [0.11, 0.22]), whose histograms are matched by construction, and recovered CycleGAN (0.74 [0.65, 0.81]) and diffusion (0.70 [0.61, 0.78]) about as well as raw slices (0.74). CycleGAN and diffusion thus transformed rather than removed the intensity site signature.

5. DISCUSSION AND RECOMMENDATIONS

Low site accuracy from a frozen probe arose from probe-specific sensitivity, from destruction, and, for every method we could test, from site information that a probe trained on outputs recovered; small change from the input was compatible with no movement toward the target. Site labels are also a confounded target: they are predictable from head geometry, which intensity harmonization should preserve, so image-probe accuracy cannot certify scanner-effect re-

moval. We recommend that evaluations (1) report several independently trained probes, including probes trained on harmonized outputs and probes restricted to the channel a method should change; (2) report each probe’s raw accuracy and a subject-level shuffled-label control; (3) measure alignment to a held-out target reference separately from change; (4) exclude target-site identities from source summaries; (5) compute paired uncertainty over subjects, not pixels or slices; (6) evaluate several draws of stochastic generators; and (7) read low site accuracy together with fidelity.

Limitations. We use one cohort, one target site and one axial slice per subject, so volumetric effects are not assessed. Without anatomical ground truth, PSNR and XCorr measure agreement with the input, not correctness. Site is confounded with geometry and demographics; our controls bound the geometric and histogram channels but do not separate scanner effects from biological site differences. Generators were not returned, and one baseline is transductive.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by ABIDE I [9]. Ethical approval was not required as confirmed by the license attached with the open access data.

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